

Technical Report

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Details on dynamical preparation of boundary modes.

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I. BACKGROUND

We consider a $2 + 1$ d dynamical quantum system subject to measurements, unitary feedback, and ancillary resets that steer arbitrary initial states toward an ensemble of 2d Chern-insulator states after a transient timescale. The dynamics consist of finite-range measurements and local unitary feedback that correct undesired occupation outcomes for truncated overcomplete Wannier (OW) modes by swapping in modes from an ancillary bath. A parameter α controls the topological phase of the dynamics.

The measurement-feedback schedule is periodic, and the dynamics can be described by a Floquet superoperator $\mathcal{F}$. The protocol therefore resembles a dynamical error-correction code: measurements are performed according to a fixed schedule, and undesired outcomes are actively corrected. We refer to this nonequilibrium system as a *dynamical Chern insulator*. For finite-range OW truncations, however, the measurements are frustrated and do not possess a common dark state. Instead, the dynamics approach a steady-state ensemble of Chern insulators that is robust to local spacetime perturbations of the circuit.

By introducing a spatially dependent α , we can create a topological interface between two distinct phases. We show that the dynamics can prepare a topological boundary mode at the interface and that the mode is robust to local perturbations of the circuit. Furthermore, the domain-wall adaptive circuit shows evidence of chiral, critical, topologically protected boundary modes at the interface, consistent with the expected universal conformal-field-theory description of the boundary physics. We study the critical phenomena of the steady-state ensemble and its dynamics.

II. ADAPTIVE CIRCUIT CONSTRUCTION

A. Parent Hamiltonian and Overcomplete Wannier Functions

1. A Chern Insulator in 2d and Its Stabilizer Formalism

We present the stabilizer formalism for a 2d Chern-insulator state $|\text{CI}\rangle$, setting the stage for the construction of the topological adaptive circuit that targets this state. Concretely, we focus on the ground state $|\text{CI}\rangle$ of a noninteracting two-band parent Hamiltonian $\hat{H}_{\text{CI}}$. In real space, the system is defined on a 2d square lattice with two fermion modes $\hat{c}_{\mathbf{r},1}$ and $\hat{c}_{\mathbf{r},2}$ per unit cell. The unit-cell coordinate $\mathbf{r} = (x, y) \in \mathbb{Z}^2$ is a pair of integers. For convenience, we start with the momentum-space resolved parent Hamiltonian of the target state $|\text{CI}\rangle$ as

$$\hat{H}_{\text{CI}} = \int \frac{d^2\mathbf{k}}{(2\pi)^2} \hat{c}^\dagger(\mathbf{k}) [\mathbf{n}(\mathbf{k}) \cdot \boldsymbol{\sigma}] \hat{c}(\mathbf{k}), \quad (1)$$

with

$$\mathbf{n}(\mathbf{k}) = (\sin k_x, \sin k_y, -\alpha - \cos k_x - \cos k_y). \quad (2)$$

$\boldsymbol{\sigma} = (\sigma^x, \sigma^y, \sigma^z)$ is the vector of Pauli matrices, and $\hat{c}(\mathbf{k}) = (\hat{c}_1(\mathbf{k}), \hat{c}_2(\mathbf{k}))^\top$ is the two-component momentum-space fermion operator. Here, $\alpha \in \mathbb{R}$ tunes the target state $|\text{CI}\rangle$. The parent Hamiltonian $\hat{H}_{\text{CI}}$ gives rise to two single-particle bands, and $|\text{CI}\rangle$ is the insulating ground state with the lower band fully occupied and the upper band empty. The band gap is finite for $\alpha \neq 0, \pm 2$, as we assume unless stated otherwise. The state $|\text{CI}\rangle$ has Chern number 1 for $\alpha \in (-2, 0)$, -1 for $\alpha \in (0, 2)$, and 0 for $|\alpha| > 2$.

The single-particle projectors onto the upper and lower bands of $\hat{H}_{\text{CI}}$ are denoted by P_+ and P_- , respectively:

$$P_\pm(\mathbf{k}) = \frac{1}{2} \left[\mathbf{1} \pm \frac{\mathbf{n}(\mathbf{k})}{|\mathbf{n}(\mathbf{k})|} \cdot \boldsymbol{\sigma} \right]. \quad (3)$$

Using this projector, we can write down the (normalized) localized fermion operators $\hat{\chi}_{\mathbf{r},A/B,\pm}$ whose single-particle OW wavefunction forms an overcomplete basis of the upper (+) and the lower (−) bands:

$$\hat{\chi}_{\mathbf{r},A/B,\pm} \propto \int \frac{d^2\mathbf{k}}{(2\pi)^2} e^{i\mathbf{k} \cdot \mathbf{r}} \tau_{A/B}^\dagger P_\pm(\mathbf{k}) \hat{c}(\mathbf{k}). \quad (4)$$

where $\tau_A = (1, 1)^\top / \sqrt{2}$ and $\tau_B = (1, -1)^\top / \sqrt{2}$. Up to the normalization factors implicit in Eq. (4), the fermion operators $\hat{\chi}_{\mathbf{r},A/B,\pm}$ are the projections of the

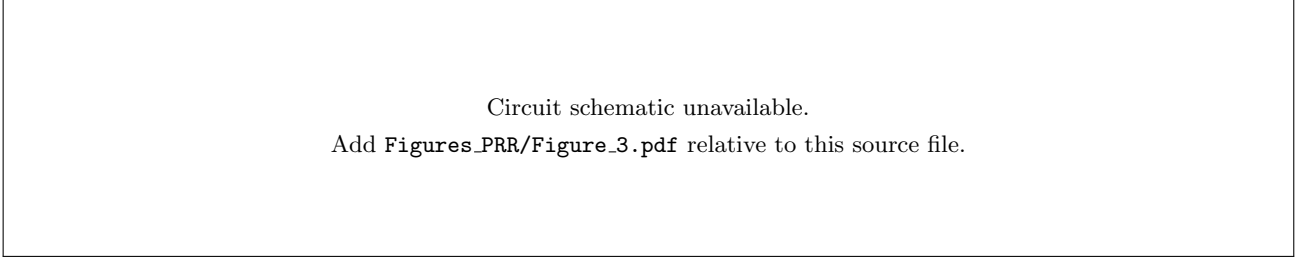


FIG. 1. Schematic for the topological adaptive circuit. The physical system (red on the top layer) and the ancillary system (blue on the bottom layer) are both 2d square lattices of the same size with two fermion modes per unit cell. In each cycle, the circuit first measures the localized number operators $\hat{N}_{\mathbf{r},\nu,\pm}$ in the physical system and applies feedforward operations to correct the measurement outcomes (red boxes). The circuit then redistributes the particles in the ancillary system (blue box) to prepare for the next cycle.

local fermion operators $\hat{c}_{\mathbf{r},A} = (\hat{c}_{\mathbf{r},1} + \hat{c}_{\mathbf{r},2})/\sqrt{2}$ and $\hat{c}_{\mathbf{r},B} = (\hat{c}_{\mathbf{r},1} - \hat{c}_{\mathbf{r},2})/\sqrt{2}$ onto the upper (+) and lower (−) bands of $\hat{H}_{\text{CI}}$. In principle, any choice of two orthogonal vectors $\tau_{A/B}$ suffices for our construction. For example, one may instead choose $\tau'_A = (1, 0)^\top$ and $\tau'_B = (0, 1)^\top$; the corresponding localized OW modes are then the projections of $\hat{c}_{\mathbf{r},1}$ and $\hat{c}_{\mathbf{r},2}$. With our chosen $\tau_{A/B}$, the OW-mode normalization factors remain $O(1)$ for all α , which is convenient for subsequent numerical studies of the adaptive dynamics.

The single-particle OW state $|W_{\mathbf{r},A/B,\pm}\rangle$ associated with the localized fermion operator $\hat{\chi}_{\mathbf{r},A/B,\pm}$ has the wavefunction:

$$W_{\mathbf{r},A/B,\pm}(\mathbf{r}', \mu') \propto \int \frac{d^2\mathbf{k}}{(2\pi)^2} e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}')} \left[\tau_{A/B}^\dagger P_\pm(\mathbf{k}) \right]_{\mu'}, \quad (5)$$

where $[\dots]_\mu$ denotes the μ th component of the OW function. This wavefunction is exponentially localized around $\mathbf{r}$ whenever the band gap in $\hat{H}_{\text{CI}}$ is nonzero, i.e., for $\alpha \neq 0, \pm 2$, leading to a localization length tunable by α .

The set of OW states $\{|W_{\mathbf{r},\nu,\pm}\rangle\}_{\mathbf{r} \in \mathbb{Z}^2, \nu=A,B}$ forms an overcomplete localized basis for the upper (+) or lower (−) band of $\hat{H}_{\text{CI}}$, although neither $\{|W_{\mathbf{r},A,\pm}\rangle\}_{\mathbf{r} \in \mathbb{Z}^2}$ nor

$\{|W_{\mathbf{r},B,\pm}\rangle\}_{\mathbf{r} \in \mathbb{Z}^2}$ is complete on its own. This overcompleteness guarantees that $|\text{CI}\rangle$ is the unique many-body state satisfying both stabilizing conditions below. Distinct OW stabilizers need not commute; the OW states form an overcomplete collection, not an orthonormal Wannier basis. These facts are crucial for convergence of the topological adaptive circuit.

The stabilizing condition for the Chern insulator state $|\text{CI}\rangle$ is given by

$$\hat{N}_{\mathbf{r},\nu,-}|\text{CI}\rangle = |\text{CI}\rangle, \quad \forall \mathbf{r} \in \mathbb{Z}^2, \quad \nu = A, B, \quad (6)$$

$$(\hat{\mathbf{1}} - \hat{N}_{\mathbf{r},\nu,+})|\text{CI}\rangle = |\text{CI}\rangle, \quad \forall \mathbf{r} \in \mathbb{Z}^2, \quad \nu = A, B, \quad (7)$$

where the number operators $\hat{N}_{\mathbf{r},A/B,\pm}$ are those of the normalized localized OW modes $\hat{\chi}_{\mathbf{r},\nu,\pm}$:

$$\hat{N}_{\mathbf{r},\nu,\pm} \equiv \hat{\chi}_{\mathbf{r},\nu,\pm}^\dagger \hat{\chi}_{\mathbf{r},\nu,\pm}, \quad (8)$$

for $\nu = A, B$. For $\alpha \neq 0, \pm 2$, when the band gap in $\hat{H}_{\text{CI}}$ is nonzero, each $\hat{N}_{\mathbf{r},A/B,\pm}$ is exponentially localized around $\mathbf{r}$. The localization length diverges as the gap closes. At finite localization length, the resulting adaptive circuit respects locality and steers the system toward the target Chern-insulator state $|\text{CI}\rangle$. The parameter α thereby interpolates between class-A many-body evolution operator (mEO) dynamics with different steady-state Chern numbers.